Spatial Transcriptomics Cheatsheet

The tables below consist of valuable functions or commands that will help you through this module.

Each table represents a different library/tool and its corresponding commands.

> Please note that these tables are not intended to tell you all the information you need to know about each command.
>
> The hyperlinks found in each piece of code will take you to the documentation for further information on the usage of each command. Please be aware that the documentation will generally provide information about the given function's most current version (or a recent version, depending on how often the documentation site is updated). This will usually (but not always!) match what you have installed on your machine. If you have a different version of R or other R packages, the documentation may differ from what you have installed.

Table of Contents

`SpatialExperiment`

Read the `SpatialExperiment` package documentation (and e-book) and a vignette on its usage.

`SpatialExperiment` objects are based on `SingleCellExperiment` objects. There are many functions that are shared between the two, with equivalent usage. See this `SingleCellExperiment` vignette for example usage of shared functions. You can also refer to the Data Lab cheatsheets for our "Introduction to scRNA-seq" (`scRNA-seq-cheatsheet.pdf`) and "Advanced scRNA-seq" (`scRNA-seq-advanced-cheatsheet.pdf`) training modules.

Library/Package	Piece of Code	What it's called	What it does
`SpatialExperiment`	`SpatialExperiment()`	Spatial Experiment	Creates a `SpatialExperiment` object
`SpatialExperiment`	`spatialCoords()`	Spatial Experiment coordinates	Extracts spatial coordinates of spots in the `SpatialExperiment` object
`SpatialExperiment`	`imgData()`	Spatial Experiment images	View the image data associated with a `SpatialExperiment` object

`VisiumIO`

Read the R package `VisiumIO` documentation and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`VisiumIO`	`TENxVisium()`	10x Visium	Constructor function for importing 10x Visium data
`VisiumIO`	`import()`	Import	Loads 10x Visium data defined with the `TENxVisium()` constructor function

`ggspavis`

Read the `ggspavis` package documentation and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`ggspavis`	`plotCoords()`	Plot coordinates	Creates a spot plot showing spatial locations in x/y with options for spot annotation
`ggspavis`	`plotVisium()`	Plot Visium	Creates a spot plot for Visium data specifically showing spatial locations in x/y with options for spot annotation as well as an option to show the spatial image, e.g., the H&E image

`SpotSweeper`

Read the `SpotSweeper` package documentation and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`SpotSweeper`	`localOutliers()`	Local outliers	Detect local outlier spots in a `SpatialExperiment` based on a given quality control metric
`SpotSweeper`	`plotQCmetrics()`	Plot QC metrics	Create a spot plot highlighting outliers detected with `SpotSweeper::localOutliers()`

`scuttle`

Read the `scuttle` package documentation, and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`scuttle`	`computeLibraryFactors()`	Compute Library Factors	Returns a numeric vector of computed size factors for each spot (or cell) stored in a `SpatialExperiment` (or `SingleCellExperiment`) object. The size factor is computed as the library size of each spot/cell after scaling them to have a mean of 1 across all spots/cells
`scuttle`	`logNormCounts()`	Normalize log counts	Returns the `SpatialExperiment` (or `SingleCellExperiment`) object with normalized expression values for each spot (cell), using the size factors stored in the object

`patchwork`

Read the `patchwork` package documentation, and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`patchwork`	`wrap_plots()`	Wrap plots	Wrap multiple `ggplot2` objects into a single multi-panel plot
`patchwork`	`+`	Plot arithmetic	Place `ggplot2` objects side-by-side by adding them together with a `+`. The `patchwork()` library must be loaded to use this symbol with plots
`patchwork`	`/`	Plot arithmetic	Stack `ggplot2` objects on top of one another "dividing" them with a `/`. The `patchwork()` library must be loaded to use this symbol with plots